Lesson Notes

MODULE 2 | LESSON 4

FACTOR MODELS IN PORTFOLIO THEORY

Reading Time	120 minutes
Prior Knowledge	Markowitz mean-variance portfolio optimization, regression models using both ordinary least squares and weighted or generalized least squares, multi-step estimation methods.
Keywords	Macroeconomic factors, fundamental factors, statistical (latent) factors, factor-mimicking portfolio, return-based style analysis

In the previous lesson, we looked at what changes are necessary when a portfolio contains assets denominated in foreign currencies and whether one should hedge one's exposure via derivatives. In this lesson, the emphasis shifts back to the original problem of min-variance optimization. Estimating all the parameters necessary is hard, often impossible, and without parameters, we cannot perform any optimization in a reliable manner. Factor models allow us to reduce the parameters to be estimated considerably. And, factors can have a life of their own and we will look at some of the possible applications.

1. Generalities

By introducing structural multi-factor risk models, it is possible to shift Markowitz's approach to portfolio construction toward a two-stage approach.

In the first stage, individual assets are grouped into classes and the optimization is performed at the level of asset classes. Once asset allocation has been completed, managers specializing in selecting securities step in.

While asset classes are usually small (20–30, occasionally more), the selection of individual securities can require the analysis of thousands of individual items. When there are so many units to analyze, it becomes impossible to estimate all the parameters one needs from the data.

This problem led to the introduction of structural multi-factor risk models, which attempt to identify a certain (relatively small) number of common factors that are able to explain the returns of individual securities.

Factors and asset classes are different. Asset classes are usually well-defined, stable, and directly investable. On the other hand, factors represent exposure to risk and, in general, are not directly investable. Some of them are even latent (hidden). The process of identifying and measuring factors is both art and science. We can distinguish between three types of factors:

- observable economic/financial time series (**macroeconomic factors**);
- created from observable asset characteristics (**fundamental factors**);
- not observable and extracted from asset returns (**statistical or latent factors**).

The first two groups can comprise economic variables like inflation, GNP growth, accounting ratios like P/BV (price-to-book value), EPS or EPS growth, etc. On the other hand, the third group contains those factors that are like principal components and that may or may not have an obvious physical meaning.

To stress the point about the difference between asset classes and factors, one should keep in mind that factors appear to have strong diversification properties because they often have low correlations with each other and sometimes even negative correlations with one another. On the other hand, asset classes usually exhibit strong positive correlations.

One question will have to be answered: whether factors really improve diversification or not. Nevertheless, by constructing portfolios based on risk factors, it is possible to remove what are sometimes artificial constraints in the definition of asset classes and managers can focus on risk drivers that affect their entire portfolio.

2. Factor Models Setup

This section is important for becoming acquainted with the necessary statistical language, which will later allow us to estimate loading parameters and factors themselves. Also, there are some important nuances that need to be well understood.

The three types of factors listed above can be treated just like the same object. We can specify factor models in three different ways.

The general form is:

$$R_{it} = \alpha_i + \sum_{k=1}^{K} \beta_{ki} f_{kt} + \epsilon_{it} = \alpha_i + \beta_i^T f_t + \epsilon_{it}$$

where:

- R_{it} is the return on asset i for $i = 1, 2, \ldots, M$ in time period $t = 1, 2, \ldots, T$. The returns can be real returns if there are no risk-free assets and excess returns when a risk-free asset is available;
- f_{kt} is the k-th factor $k = 1, 2, \ldots K$ at time t;
- β_{ik} is the sensitivity beta (also called loading) for asset i with respect to factor k;
- ϵ_{it} is the asset specific (residual factor).

Some assumptions are needed:

- the factors need to be stationary and their unconditional moments are given by

$$E[f_t] = \mu_f, \;\; \text{cov}(f_t) = \Sigma_f.$$

- the error terms are uncorrelated with each one of the common factors, i.e.,

$$\text{cov}(f_{kt}, \epsilon_{it}) = 0 \;\; k = 1, 2, \ldots, K, i = 1, 2, \ldots, M, \quad \text{and} \quad t = 1, 2, \ldots, T.$$

- Finally, the error terms are serially uncorrelated, but contemporaneously correlated across assets, that is

$$\text{cov}(\epsilon_{it}, \epsilon_{js}) = \begin{cases} \sigma_i^2 & \text{for } i = j, t = s; \\ 0 & \text{otherwise .} \end{cases}$$

Furthermore, we will use the convention that

$$R_t = \begin{pmatrix} R_{1t} \\ R_{2t} \\ \vdots \\ R_{Mt} \end{pmatrix} \;\; t = 1, 2, \ldots, T$$

represents a vector that is the **cross section** of all assets' returns at time t.

On the other hand, if we wrote

$$R_i = \begin{pmatrix} R_{i1} \\ R_{i2} \\ \vdots \\ R_{iT} \end{pmatrix} \;\; i = 1, 2, \ldots, M$$

that should be recognized as the **time series** of the returns of asset i over the time framework used.

Finally, all assets' returns over the full time period are represented as the matrix

$$R = \begin{pmatrix} R_{11} & R_{21} & \cdots & R_{M1} \\ R_{12} & R_{22} & \cdots & R_{M2} \\ \vdots & \vdots & \ddots & \vdots \\ R_{1T} & R_{2T} & \cdots & R_{MT} \end{pmatrix}.$$

Hence, we examine factor models through 3 different lenses:

- time series regressions;
- cross-section regressions;
- multivariate regressions.

As a **cross-section regression**, this model looks like:

$$R_t = \alpha + Bf_t + \epsilon_t \;\; t = 1, 2, \ldots, T.$$

In this setup:

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$$R_t = \begin{pmatrix} R_{1t} \\ R_{2t} \\ \vdots \\ R_{Mt} \end{pmatrix}, \ \alpha = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_M \end{pmatrix}, \ \epsilon_t = \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \\ \vdots \\ \epsilon_{Mt} \end{pmatrix},$$

$$f_t = \begin{pmatrix} f_{1t} \\ f_{2t} \\ \vdots \\ f_{Kt} \end{pmatrix}, \ B = \begin{pmatrix} \beta_{11} & \beta_{12} & \cdots & \beta_{1K} \\ \beta_{21} & \beta_{22} & \cdots & \beta_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{M1} & \beta_{M2} & \cdots & \beta_{MK}. \end{pmatrix}$$

In particular, we have that model is *heteroskedastic* since

$$E[\epsilon_t \epsilon_t^T] = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_M^2).$$

The model can also be written as a **time series regression** in the form

$$R_i = 1_T \alpha_i + F\beta_i + \epsilon_i \ \ i = 1, 2, \ldots, M.$$

In this case,

$$R_i = \begin{pmatrix} R_{i1} \\ R_{i2} \\ \vdots \\ R_{iT} \end{pmatrix}, \ 1_T = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \ \beta_i = \begin{pmatrix} \beta_{i1} \\ \beta_{i2} \\ \vdots \\ \beta_{iK} \end{pmatrix},$$

$$F = \begin{pmatrix} f_{11} & f_{21} & \cdots & f_{K1} \\ f_{12} & f_{22} & \cdots & f_{K2} \\ \vdots & \vdots & \ddots & \vdots \\ f_{K1} & f_{K2} & \cdots & f_{KT}. \end{pmatrix}, \ \epsilon_i = \begin{pmatrix} \epsilon_{i1} \\ \epsilon_{i2} \\ \vdots \\ \epsilon_{iT} \end{pmatrix},$$

and in this case $E[\epsilon_i \epsilon_i^T] = \sigma_i^2 I_T$ with I_T the $T \times T$ unit matrix, i.e., the error covariance is *homoskedastic*.

For the multivariate version there are at least 3 different ways to write it. The one we propose here is perhaps the most suitable for direct estimation:

$$R = X\Delta + E$$

where

$$R = \begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ R_M \end{pmatrix}, \ Q = \begin{pmatrix} 1 & f_{11} & f_{21} & \cdots & f_{K1} \\ 1 & f_{12} & f_{22} & \cdots & f_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & f_{1T} & f_{2T} & \cdots & f_{KT} \end{pmatrix}, \ X = \begin{pmatrix} Q & \emptyset & \cdots & \emptyset \\ \emptyset & Q & \cdots & \emptyset \\ \vdots & \vdots & \ddots & \vdots \\ \emptyset & \emptyset & \cdots & Q \end{pmatrix}, \ E = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_M \end{pmatrix}$$

where R_i and ϵ_i were defined above and $\emptyset$ is a$T \times (K+1)$ matrix of zeroes. Finally,

$$\Delta = (\alpha_1, \beta_{11}, \beta_{21}, \cdots, \beta_{K1}, \alpha_2, \beta_{12}, \beta_{22}, \cdots, \beta_{K2}, \cdots, \alpha_M, \beta_{1M}, \ \beta_{2M}, \cdots, \beta_{KM})^T$$

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and

$$\text{cov}(E) = \begin{pmatrix} \sigma_1^2 I_T & \emptyset & \cdots & \emptyset \\ \emptyset & \sigma_2^2 I_T & \cdots & \emptyset \\ \vdots & \vdots & \ddots & \vdots \\ \emptyset & \emptyset & \cdots & \sigma_M^2 I_T \end{pmatrix} \equiv D.$$

Note that $\text{cov}(E)$ is *homoskedastic, within each asset*, and the structure of the regression allows us to estimate the parameters Δ in an efficient manner using OLS for the individual time series regressions for each of the M assets.

However, computationally speaking, it might be convenient to use the multivariate regression setup. The matrix D is heteroskedastic, and in that case, we would need to use generalized or weighted least squares (GSL/WLS).

3. Factor Models and Covariance Structure

This section addresses the problem of estimation mentioned in the introduction.

In a traditional mean-variance optimization framework with M assets, we need to estimate many parameters: M returns,M variances, and $M(M-1)/2$ covariances: a total of $2M + M(M-1)/2$ parameters. If $M = 10$ that makes 65 parameters; if $M = 100$ that would be 5,150 parameters; for $M = 500$ it amounts to 125,750 parameters.

However, the problem does not lie so much in the number of parameters per se, but the fact that, using the traditional timeframes adopted for this type of analysis, the standard deviations of our estimates for the means are larger than the estimates themselves. That makes the estimates useless (**historical blur**) and could mislead a portfolio manager into a false sense of security about their optimized portfolio.

The only way around the issue would be to increase the number of data points, i.e., considering longer periods of time by going back 20, 30, 50, or more years, instead of the more traditional 2, 3, or 5. While this may not be enough to solve the problem, it is also not desirable since, over a very long period of time, we cannot assume that the parameters have not changed. When working with high-frequency data (as with Bitcoin and other cryptocurrencies) we would have more data, but that data could still contain a lot of noise and not allow for precise estimates.

Factor models can be of assistance. In fact, if we use the cross-section setup:

$$R_t = \alpha + Bf_t + \epsilon_t \;\; t = 1, 2, \ldots, T,$$

then under the assumptions of the multifactor model, the $M \times M$ covariance matrix of asset returns has the form

$$\text{cov}(R_t) = \Omega = B\Sigma_f B^T + \Sigma_\epsilon, \;\; \Sigma_\epsilon = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_M^2).$$

This last result implies that

$$\text{var}(R_{it}) = \beta_i^T \Sigma_f \beta_i + \sigma_i^2, \;\; \text{cov}(R_{it}, R_{jt}) = \beta_i^T \Sigma_f \beta_j.$$

Notice that, given our assumption about the structure of the model and the stationarity, these estimates do not depend on t as long as the loading factors,B are time invariant (which we are assuming). Even accounting for the fact that one has to estimate B and Σ_f and the σ_i^2's, the amount of parameters is considerably smaller than using the direct approach.

Hence, the first contribution of factor models is that, under certain circumstances, they provide a way to alleviate the historical blur issue in estimating the parameters for a mean-variance optimization program.

Example (Porfolio factor model). Let $w = (w_1, \ldots, w_M)^T$ be the vector of weights that identify a porfolio of M assets. Then, the return on the portfolio is given by

$$R_{pt} = w^T R_t$$

If we use the (cross section) factor model for R_t, i.e.,

$$R_t = \alpha + Bf_t + \epsilon_t,$$

then

$$R_{pt} = w^T \alpha + w^T B f_t + w' \epsilon_t$$

which shows how the return on the portfolio is related to the factors chosen. In particular, $\beta_p^T = w^T B$ represents the loading/sensitivities of the portfolio with respect to the factors selected. In addition,

$$\mathbf{var}(R_{Pt}) = w^T B \Sigma_f B^T w + w^T \Sigma_\epsilon w.$$

4. Factors and Estimation of Coskweness and Cokurtosis Matrices.

If we have a model like

$$R_t = a + Bf_t + \epsilon_t,$$

where B as before is the matrix of factor loadings and F the matrix of factors, we have just seen how that allows us to estimate the covariance matrix using fewer parameters. What happens to the coskewness and cokurtosis matrices? Well, there are formulas linking the matrices B and F to the coskewness and cokurtosis matrices. Using the same symbols we used in Lesson 2, these are:

$$M_3 = BM_{3,f}(B^T \otimes B^T) + \Omega$$

$$M_4 = BM_{4,f}(B^T \otimes B^T \otimes B^T) + \Gamma.$$

Here $M_{3,f}$ and $M_{4,f}$ are the coskweness and cokurtosis matrices for the factors.Ω and Γ here are special matrices. Specifically,Ω is an $N \times N^2$ matrix of zeroes, except for the i, j elements where $j = (i-1)N + h$ (recall that for coskewness we need three indices) corresponding to the expected third moment of the terms ϵ_t. Γ is more complicated, unfortunately. The definitions and the proofs of these decompositions are provided in a paper by Boudt et al. listed among the references, but which is not open access, unfortunately. A link to a paper, which is freely available, is provided in the references (albeit the code therein is in R and not in Python).

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5. Macroeconomic Factor Models

Arguably, the most famous example of a macroeconomic factor model is the **single factor model** introduced by Sharpe in 1970. This model is:

$$R_{it} = \alpha_i + \beta_i R_{Mt} + \epsilon_{it}, \;\; i = 1, 2, \ldots, N; t = 1, 2, \ldots, T$$

where R_{Mt} is the return of the market index in excess of the risk-free rate and it is called the **market risk factor.**

*(**Note:** We change the index M to N to avoid confusion with the market risk factor).*

In this case,R_{it} are the returns of asset i in excess of the risk-free rate as well. Then, for instance, we could use this model to test if the CAPM hold ($\alpha_i = 0$ for each i). In this setting we have a single factor i.e.$f_{1t} = R_{Mt}$. It is easy to see that in this particular model:

$$\sigma_M^2 = \mathbf{var}(R_{Mt}), \;\; \beta^T = (\beta_1, \ldots, \beta_N)^T, \;\; \mathbf{cov}(R_t) = \sigma_M^2 \beta\beta^T + \Psi$$

with $\Psi = \mathbf{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_N^2)$.
Regarding the estimation of this model, if we use the time series regression models for each asset i, these models satisfy the homoskedasticity assumption, and thus, OLS are the best linear unbiased estimates. Hence, let $\widehat{\alpha}_i, \widehat{\beta}_i, i = 1, 2, \ldots, N$ be the OLS estimates.

Then, we can estimate the residuals as

$$\hat{\epsilon}_{it} = R_{it} - \widehat{\alpha}_i - R_{Mt}\widehat{\beta}_i$$

and

$$\widehat{\sigma}_i^2 = \frac{1}{T-2}\sum_{t=1}^{T} \hat{\epsilon}_{it}^2, \;\; i = 1, 2, \ldots, N.$$

This allows us to obtain $\widehat{\Psi} = \mathbf{diag}(\widehat{\sigma}_1^2, \ldots, \widehat{\sigma}_N^2)$. Finally,

$$\widehat{\sigma}_M^2 = \frac{1}{T-1}\sum_{t=1}^{T}(R_{Mt} - \bar{R}_M)^2$$

and the unconditional covariance for R_t is then given by

$$\widehat{\mathbf{cov}(R_t)} = \widehat{\sigma}_M^2 \widehat{\beta}\widehat{\beta}^T + \widehat{\Psi}.$$

The extension to the multivariate case is straightforward as one can estimate M time series regressions:

$$\begin{pmatrix} \widehat{\alpha}_i \\ \widehat{\beta}_{1i} \\ \widehat{\beta}_{2i} \\ \vdots \\ \widehat{\beta}_{Ki} \end{pmatrix} = (Q^T Q)^{-1} Q R_i \;\; i = 1, 2, \ldots, M,$$

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and where the matrix q was defined earlier.

The estimated residuals can be obtained as usual

$$\hat{\epsilon}_{it} = R_{it} - \widehat{\alpha}_i - Q\widehat{\beta}_i$$

so that

$$\widehat{\sigma}_i^2 = \frac{1}{T-K-1}\sum_{t=1}^{T}\hat{\epsilon}_{it}^2.$$

Finally,

$$\widehat{\Sigma}_f = \frac{1}{T-1}\sum_{t=1}^{T}(f_t - \bar{f})(f_t - \bar{f})^T \quad \text{and}$$

$$\widehat{\Sigma} = \widehat{B}\widehat{\Sigma}_f\widehat{B}^T + \widehat{\Psi}.$$

None of these regressions present issues since the factors are observable.

6. Fundamental Factors

The goal of fundamental factor models is to identify features that are relevant to forecasting potential return and risk for securities. Some of these factors are microeconomic (industry, earnings growth, price-to-book value, etc). Fundamental models decompose risk using a few easily understood elements. Intuitively appealing, fundamental models are ideal for performance attribution and portfolio risk decomposition. They are also supported by decades of academic research in asset pricing theory. However, since fundamental models are defined in terms of known factors, they can become less accurate when new, and until then unknown, factors emerge as relevant.

Arguably, one of the most famous fundamental models is the **Fama and French Three Factor Model**. This model uses three factors:

$$R_{it} - r_{ft} = \alpha_{it} + \beta_1(R_{Mt} - r_{ft}) + \beta_2 SMB_t + \beta_3 HML_t + \epsilon_{it}$$

where:

- R_{it} is the return of asset i at time t;
- r_{ft} is the risk-free rate of return at time t;
- R_{Mt} is the return on the market portfolio;
- $R_{it} - r_{ft}$ and $R_{Mt} - r_{ft}$ are excess returns on asset i and the market portfolio, respectively;
- $R_{Mt} - r_{ft}$ is the market premium (momentum factor);
- SMB_t is the size premium (small minus big). This time series represents the historical excess returns of small companies over large cap companies (*size factor*);
- HML_t is the value premium (high minus low). This time series contains the historical excess returns of value stocks vs. growth stocks. A stock is a value stock if it has a high book-to-price ratio, and it is a growth stock if it has a low book-to-price ratio (*value factor*) instead.

The values for SMB and HML time series and explanations about the methodology are found on Kenneth French's website at Dartmouth University (Kenneth French's datasets).

Example. As an application, we can estimate a 3-factor model for Apple stock during the period January 2017 to February 2022.

The OLS regression is shown below. We notice that:

- the α is statistically not significant, which is what we should expect if the asset pricing theory (APT) model holds.
- The coefficient of the market risk $\beta_1 = 1.259$ is like a beta (it would be if we had estimated the model without the other regressors) and shows that the market returns affect the returns of Apple stock as well (momentum);
- $\beta_2 = -0.4903$ tells us that the size factor has a negative impact, i.e., Apple, being a large cap company, is actually less risky and thus gets a discount in terms of returns. However,β_2 is not statistically significant at the 90%-confidence level;
- $\beta_3 = -0.5313$ is statistically significant and the value factor has a negative impact. In other words, Apple qualifies like a growth stock rather than a value stock.

Figure 1: Estimated 3-Factor Model for Apple during the Period January 2017 to February 2022

OLS Regression Results

Dep. Variable:	excess_rtn	R-squared:	0.477
Model:	OLS	Adj. R-squared:	0.449
Method:	Least Squares	F-statistic:	17.04
Date:	Wed, 02 Mar 2022	Prob (F-statistic):	5.50e-08
Time:	11:06:35	Log-Likelihood:	82.465
No. Observations:	60	AIC:	-156.9
Df Residuals:	56	BIC:	-148.6
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	0.0140	0.009	1.622	0.110	-0.003	0.031
mkt	1.2590	0.185	6.812	0.000	0.889	1.629
smb	-0.4903	0.316	-1.550	0.127	-1.124	0.143
hml	-0.5313	0.206	-2.580	0.013	-0.944	-0.119

Omnibus:	6.222	Durbin-Watson:	2.204
Prob(Omnibus):	0.045	Jarque-Bera (JB):	6.386
Skew:	-0.443	Prob(JB):	0.0410
Kurtosis:	4.330	Cond. No.	39.7

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Fama and French also have models with 4 or 5 factors. Chen et al. published a model with 5 factors as well.

- industrial production,
- changes in the risk premium,
- twists in the yield curve,
- measures of unanticipated inflation,
- changes in inflation during high volatile periods for these variables.

This model, unlike Fama and French, contains only observed macro-economic factors. Fung and Hsieh proposed a risk-factor model with 8 factors, which is applied to hedge funds. Barra Inc.'s Global Equity Model (GM3) uses 34 industry factors based on GICS. Axioma provides world-wide

equity factor risk models. Other factor models are supplied by Morningstar and Bloomberg, etc. These indices are mostly not free as it is a lot of work to update the time series, rebalance them, etc.

The Barra model incorporates two types of factors: style and industry. Barra style factors include momentum, size, earning yield, residual volatility, growth, dividend yield, book-to-price, leverage, liquidity, etc.

The industry factors are not necessarily what one thinks, at least not completely, as defining an industry is somewhat arbitrary. We will discuss this in the next section. However, their model looks like this:

$$R_{it} = \alpha_i f_{ct} + \sum_{k=1}^{K} \beta_{ik} f_{kt}^{(I)} + \sum_{h=1}^{H} \beta_{ih} f_{ht}^{(S)} + \epsilon_{it}, \;\; i = 1, 2, \ldots, N$$

where the $f_k^{(I)}$ factors are industry sectors and the $f_h^{(s)}$ factors are style factors. In this case, the loading matrix vector β_i is the exposure of asset i to the industry factors. If stock i is in industry g, then $\beta_{ig} = 1$ and $\beta_{ig} = 0$ when $i \neq g$. f_c is a country factor, in which case every stock has an exposure of 1 to its country factor.

7. Statistical Factors

The main difference among statistical/latent factors and the previous two groups is that statistical factors are not observable. And it is a big difference!

To understand the big change, we can start with a relatively easy situation like an industry factor model in which we assume that there are H industries. We could choose H as large as the number of stocks, but that would defeat the purpose, or we could choose H as a number of, say, five industries. Now, five may not be enough, but the emphasis here is on the nuances of the methodology and not its accuracy. Let's say that we have N risky assets and $H < N$ industries. Then, an industry factor model looks like this:

$$R_{it} = \gamma_{i1} f_{1t} + \gamma_{i2} f_{2t} + \ldots + \gamma_{iH} f_{Ht} + \epsilon_{it}, \;\; i = 1, 2, \ldots, N; t = 1, 2, \ldots T$$

where we still have $\mathbf{var}(\epsilon_{it}) = \sigma_i^2$ for $i = 1, 2, \ldots, N$, $\mathbf{cov}(f_{ht}, \epsilon_{it}) = 0$ for $h = 1, \ldots, H$ and $j = 1, \ldots N$, and $\mathbf{cov}(\epsilon_{it}, \epsilon_{is}) = 0$ for $s \neq t$. However, now we have additional assumptions:

$$\gamma_{ih} = \begin{cases} 1 & \textbf{if asset i is in industry h;} \\ 0 & \textbf{otherwise.} \end{cases}$$

Notice the key difference: _We are assuming that the γ_{ik} are the known parameters, while the factors are the unknown_. Hence, in this type of model, we will estimate the factors and not the γ's. Also, if we assume that there are N_h assets in industry h, $h = 1, 2, \ldots, H$, then $\sum_{h=1}^{H} N_h = N$.

In addition, since the industries in our definition are mutually exclusive, we also have $\gamma_i^T \gamma_h = N_h$ if $j = h$ or zero otherwise. Unlike before, in this case, we do not know the matrix Σ_f since we do not observe the factors. Nevertheless, we need the time series of factors in order to estimate the residuals and the covariance $D = \mathbf{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2)$ which will in turn be used to compute the WLS estimates. As we are using cross sections, we need to use GLS/WLS because OLS provides

unbiased estimates but not efficient ones due to the fact that there is cross-sectional heteroskedasticity in the returns of the assets, as we learned in the setup section.

In this case, a multi-step estimation process known as **FGLS** (**feasible generalized least squares**) is required. It works like this:

Step 1: Estimate the factors using OLS which yields:

$$\hat{f}_{ht} = \frac{1}{N_h} \sum_{j=1}^{N_h} R_{jt}^{(h)} \;\; h = 1, 2, \ldots, H$$

where the notation $R_{jt}^{(h)}$ is a reminder that we are considering only the returns at time t of those stocks that belong to industry h. If we used a matrix notation, we could write the results as follows (this is just a more compact multivariate regression notation):

$$\widehat{f}_t^{OLS} = (\Gamma^T \Gamma)^{-1} \Gamma^T R_t.$$

Step 2: Estimate the factor covariance matrix. This is done from the estimates for the factors we found in Step 1. Hence

$$\widehat{\Sigma}_f^{OLS} = \frac{1}{T-1} \sum_{t=1}^{T} (\widehat{f}_t^{OLS} - \bar{f}^{OLS})(\widehat{f}_t^{OLS} - \bar{f}^{OLS})^T, \;\; \bar{f}^{OLS} = \frac{1}{T} \sum_{t=1}^{T} \widehat{f}_t^{OLS}.$$

Step 3: Once the estimates 's are available, we can compute the estimated residuals $\hat{\epsilon}_{ht}$'s. In turn, from these estimated residuals we can compute $\widehat{D}^{OLS}$ using

$$\widehat{\sigma}_i^{2,OLS} = \frac{1}{T-1} \sum_{t=1}^{T} (\hat{\epsilon}_{it}^{OLS} - \bar{\epsilon}_i^{OLS})^2 \;\; i = 1, 2, \ldots, N.$$

Step 4: This step is simple addition:

$$\widehat{\Sigma}^{OLS} = B \widehat{\Sigma}_f^{OLS} B^T + \widehat{D}^{OLS}.$$

Step 5: We now use GLS/WLS for the final step. Thus:

$$\widehat{f}_t^{GLS} = \left(B^T (\widehat{D}^{OLS})^{-1} B\right)^{-1} B^T (\widehat{D}^{OLS})^{-1} R_t, \;\; t = 1, 2, \ldots, T;$$

$$\widehat{\Sigma}_f^{GLS} = \frac{1}{T-1} \sum_{t=1}^{T} (\widehat{f}_t^{GLS} - \bar{f}^{GLS})(\widehat{f}_t^{GLS} - \bar{f}^{GLS})^T, \;\; \bar{f}^{GLS} = \frac{1}{T} \sum_{t=1}^{T} \widehat{f}_t^{GLS};$$

$$\widehat{\sigma}_i^{2,GLS} = \frac{1}{T-1} \sum_{t=1}^{T} (\hat{\epsilon}_{it}^{GLS} - \bar{\epsilon}_i^{GLS})^2 \;\; i = 1, 2, \ldots, N;$$

$$\widehat{\Sigma}^{GLS} = B \widehat{\Sigma}_f^{GLS} B^T + \widehat{D}^{GLS}.$$

For the industry factors we considered above, we could not observe the factors. However, because of the particular structure of the loadings, we knew the matrix B. The most general case is when the common factors are hidden and the loadings are not known.

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We do know the $N \times N$ covariance matrix of returns for our risky assets Σ, though. In this case, the factors must be extracted from Σ, which contains our "knowledge" about the problem. This can be done using either principal component analysis (PCA) or factor analysis. The steps using PCA are the following:

Step 1: Obtain $\widehat{\Sigma}$, the sample covariance. We have done this several times already.

Step 2: The matrix $\widehat{\Sigma}$ admits the following (spectral) representation $\widehat{\Sigma}_f = \Gamma \Lambda \Gamma^T$ where $\Lambda = \mathbf{diag}(\lambda_1, \lambda_2, \ldots, \lambda_K)$ with $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_K > 0$ the eigenvalues of $\widehat{\Sigma}_f$, and Γ the eigenvectors associated to those eigenvalues and such that $\Gamma\Gamma^T = \Gamma^T\Gamma = I_N$.

Step 3: Obtain initial estimates:

$$\tilde{\alpha}_0 = \bar{R} = \frac{1}{T} R \iota^T.$$

$$\tilde{B}_0 = \Gamma^{(k)} (\Lambda^{(k)})^{1/2}$$

where $\Gamma^{(k)}$ contains the first k columns of Γ and $\Lambda^{(k)}$ the first k columns of Λ.

$$\tilde{\Psi}_0 = \mathbf{diag}(\widehat{\Sigma}) - \mathbf{diag}(\tilde{B}_0 \tilde{B}_0^T).$$

$$\tilde{\Sigma}_0 = \tilde{B}_0 \tilde{B}_0^T + \tilde{\Psi}_0,$$

Step 4: Recompute the sample covariance matrix as $\widehat{\Sigma}^\star = \widehat{\Sigma} - \tilde{\Psi}_0$ Then, recompute the decomposition $\widehat{\Sigma}^\star = \tilde{\Gamma} \tilde{\Lambda} \tilde{\Gamma}^T$.

Step 5: Repeat step 3 using the new elements to generate $\tilde{B}_1$ and $\tilde{\Psi}_1$. Then, use these to generate $\tilde{\Sigma}_1 = \tilde{B}_1 \tilde{B}_1^T + \tilde{\Psi}_1$.

Step 6: Repeat step 5 several times to generate a sequence of estimates $\tilde{B}_u, \tilde{\Psi}_u, \tilde{\Sigma}_u$ for $u = 2, 3, \ldots$ and stop when the changes in $\tilde{\Psi}_u$ and $\tilde{\Psi}_{u+1}$ are negligible. These are the final estimates.

When we use a spectral decomposition, the eigenvectors matrix Γ contains orthogonal vectors. Each of this vectors is one component in the matrix of sample principle components because the matrix $\widehat{\Sigma}$ defined by $\Gamma \Lambda \Gamma^T$ is the covariance matrix of the sample principal component $\widehat{P} = \Gamma R$. Now, $\widehat{P}$ is an $M \times T$ matrix, and we can choose as many principal components as necessary "to do a good job," not necessarily all of them.

One question left to answer is how many factors should one use? If we consider again the spectral decomposition of Σ or $\widehat{\Sigma}$, i.e.

$$\widehat{\Sigma} = \Gamma \Lambda \Gamma^T \;\; \Gamma^{-1} = \Gamma^T,$$

the matrix Γ consists of orthonormal eigenvectors. For principal component analysis, one would take the diagonal of Λ and sort its elements into descending order. The same sorting procedure must be executed on the columns of Γ, of course. Then, the fraction of the variance explained by each principal component is given by the ratios:

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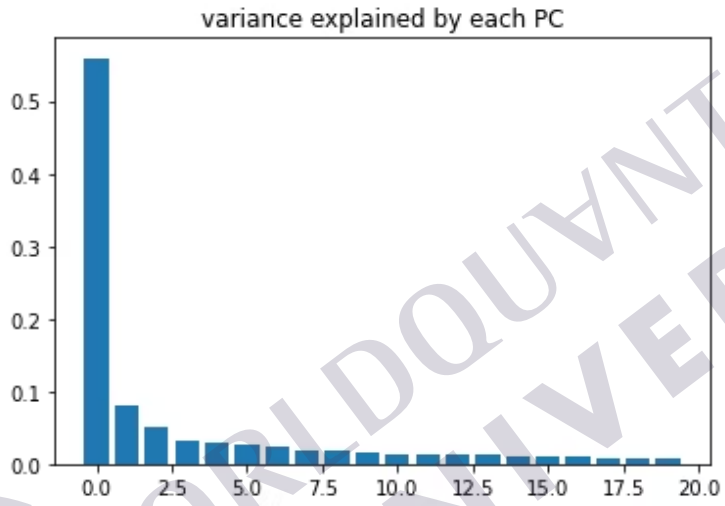
$$\frac{\lambda_i}{\sum_{i=1}^{N}\lambda_i}, \quad \text{and} \quad \frac{\sum_{i=1}^{I}\lambda_i}{\sum_{i=1}^{N}\lambda_i},$$

which represent the percentage of the total variance explained by the i-th principal component of factor returns and the ratio of the total variance explained by the first I principal components, respectively.

This can help you decide how many factors to use.

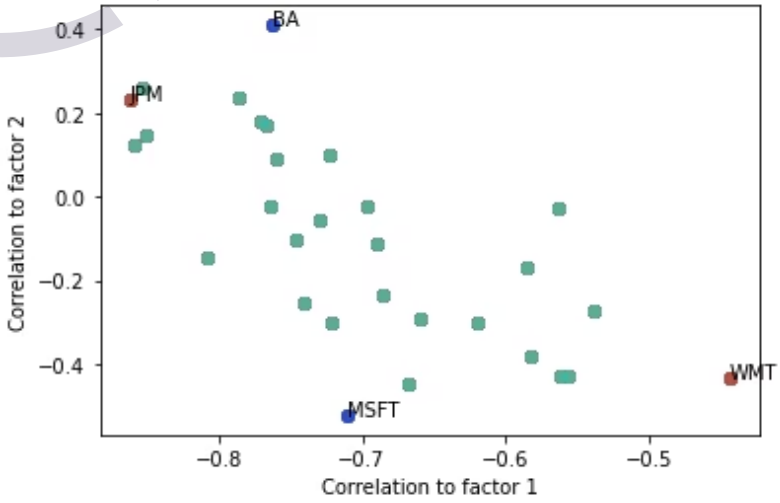
Example (Extracting principal components for the Dow Jones Industrial Average Index). If we considered the DJ Index 30 companies during the period January 2, 2020–March 4, 2022, we find that the first two principal components explain about 2/3 of the volatility as shown in figure 2.

Figure 2: Twenty Principal Components for the Dow Jones Industrial Index over the Period Jan. 2, 2020 - Mar. 4, 2022_



The next figure show the correlations of each of the 30 components of the Dow Jones Index with factor 1 and factor 2. The dots in red and blue denote the maximum and minimum correlations with the first and second factors, respectively. Principal components, when used as factors, are not easy to explain as recognizable macroeconomic or fundamental factors.

Figure 3: Correlations of Returns of Dow Jones Industrial Average 30 Individual Assets with the First and Second Principal Components Considered as Factors



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8. Factor-Mimicking Portfolio (FMP)

Assume that we have identified a factor that is relevant to us. However, the factor may not be observable in the market. How can we obtain the excess returns for it when it is not observable?

The answer is that we can create a portfolio that behaves like this factor using a group of observable ones for which returns or excess returns are available.

The idea is this: We have N risky assets for which the $N \times N$ covariance matrix of returns,Ω, is known, we also have K factors. Then, if w_k is an $N \times 1$ vector of weights, we solve the following variance minimization program:

$$\min_{w_k} \frac{1}{2} w_k^T \Omega w^T$$
$$\text{subject to } B^T w_k = \delta_k.$$

It is important to understand this part. The objective function is clear. The constraints employ B, the loading matrix we have used before. The constraints, written explicitly, look like this:

$$B^T w_k = \begin{pmatrix} \beta_{11} & \beta_{21} & \cdots & \beta_{N1} \\ \beta_{12} & \beta_{22} & \cdots & \beta_{N2} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{1K} & \beta_{2K} & \cdots & \beta_{NK} \end{pmatrix} \cdot \begin{pmatrix} w_{k1} \\ w_{k2} \\ \vdots \\ w_{kN} \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^N \beta_{i1} w_{ki} \\ \sum_{i=1}^N \beta_{i2} w_{ki} \\ \vdots \\ \sum_{i=1}^N \beta_{iK} w_{ki} \end{pmatrix} \equiv \begin{pmatrix} \delta_{k1} \\ \delta_{k2} \\ \vdots \\ \delta_{kK} \end{pmatrix} \equiv \delta_k.$$

Hence,δ_k is a$K \times 1$ vector for each value of $k = 1, 2, \ldots, K$. Now,δ_{k1} represents the loading of factor 1 for the portfolio,δ_{k2} the loading of factor 2, and so on. In particular,δ_{kk} represents the loading of factor k for the portfolio. These δ_{kh} elements are our choice. We can solve the optimization problem as we did before when dealing with Markowitz mean-variance optimization—just keep in mind that here we technically have K constraints. We will not go over the entire derivation via calculus, but the result is

$$w_k^\star = \Omega^{-1} B (B^T \Omega^{-1} B)^{-1} \delta_k.$$

The factor-mimicking portfolio weight matrix (for all K factors) is given by:

$$W_K^\star = \Omega^{-1} B (B^T \Omega^{-1} B)^{-1} \Delta_K.$$

where Δ_K is the $K \times K$ matrix $(\delta_1, \delta_2, \ldots, \delta_K)$.

Now, if we want the portfolio to mimic factor k, it seems natural that we must have $\delta_{kk} = 1$. For δ_{kh} when $h \neq k$, we do not need to assign them; nevertheless, if we set $\delta_{kh} = 0$ for $h \neq k$, we obtain a **pure factor-mimicking portfolio for factor k.**

Example: We can look at what happens if we use the formula just derived for the FMP weights under certain situations:

- $\Omega = \sigma^2 I_N$ (assets are uncorrelated and have constant variance) and $\Delta_K = I_K$, the $K \times K$ identity matrix. In this case, we find $W_K^\star = B(B^T B)^{-1}$.
- $\Omega = D \equiv \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_N^2)$, that is, the assets are uncorrelated but are allowed to have different variances and $\Delta_K = I_K$. In this case, we get $W_K^\star = D^{-1} B (B^T D^{-1} B)^{-1}$.

- In this case, we assume that Ω is unconstrained, but $\Delta_K = I_K$. Then, $W_K^\star = \Omega^1 B(B^T\Omega^{-1}B)^{-1}$.

Example. (Factors and loadings are not unique). Consider the model
$R_t = \alpha + Bf_t + \epsilon_t, \;\; t = 1, 2, \ldots, T$
(the cross-section regression notation) as done in the setup section. Now, for any $K \times K$ matrix that is invertible we can define
$f_t^\star = Hf_t$ and $B^\star = BH^{-1}$. We note the following:

$$R_t = \alpha + Bf_t + \epsilon_t = R_t = \alpha + BH^{-1}Hf_t + \epsilon_t = \alpha + B^\star f_t^\star + \epsilon_t.$$

Furthermore, we have

$$\mu_f^\star = H\mu_f, \;\; \Sigma_f^\star = \text{cov}(f_t^\star) = \text{cov}(Hf_t) = H\Sigma_f H^T.$$

The matrix H is our choice. If Σ_f is positive definite, it admits a (spectral) decomposition
$\Sigma_f = \Gamma\Lambda\Gamma^T$
where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_K)$ with $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_K > 0$ the eigenvalues of Σ_f, and Γ the eigenvectors associated with the eigenvalues. Then, if we let $H = \Gamma\Lambda^{-1/2}$, we obtain
$\Sigma_f^\star = I_K$.

Since factors and loadings are not unique, this creates some issues for their interpretation. Using the matrix H as done above is equivalent to using **orthogonal transformations**, which amount to **rotation** of the coordinates. Some rotation may be easier to interpret than others.

9. Returns-Based Style Analysis

When performing a returns-based style analysis, the goal is to identify the style of a fund through a regression of the fund's returns against a certain number of passive style indices. There are many indices available. For instance, one could check the MSCI website.

In certain applications, say that we want to understand how a portfolio manager achieves results. We can use a regression over a pre-specified period of time and see where the manager's exposure lies. This can inform us about the manager's style in investing. Comparing the regression coefficients over different periods of time, we can detect if a change in style has occurred.

Example: Let's say that we have the returns for the manager's portfolio over a period of T months, $R_{pt}, t = 1, 2, \ldots, T$. For the same period, we also have the returns for 4 indices: small-cap growth (SCG), large-cap growth (LCG), small-cap value (SCV), and large-cap value (LCV). Then, we estimate the following linear regression model:

$$R_{pt} = \alpha + \beta SCG_t + \gamma LCG_t + \delta SCV_t + \theta LCV_t + \epsilon_t, \;\; t = 1, 2, \ldots, T$$

with the constraints that $\beta + \gamma + \delta + \theta = 1$ (constrained linear regression model).
The estimated coefficients are found to be:

$$\hat{\beta} = 0.02, \hat{\gamma} = 0.05, \hat{\delta} = 0.15, \hat{\theta} = 0.78.$$

What can we say about the manager's investment style?

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- The portfolio has very little exposure to growth stocks ($\hat{\beta}$ and $\hat{\gamma}$ are very small);
- As a consequence, the manager invests mostly in value stocks. In addition, since $\hat{\theta}$ is much larger than $\hat{\delta}$, we can also state that the manager is primarily investing in large-cap value stocks.

Assume that now, it is five years later, and using new data about the manager's portfolio returns and the indices, we estimate the same regression model and this time

$$\hat{\beta} = 0.15, \hat{\gamma} = 0.30, \hat{\delta} = 0.20, \hat{\theta} = 0.35.$$

In this case, the shift in the manager's style is quite evident.

- The portfolio is now nearly split between growth and value assets;
- While large cap are still preferred, there is definitely less emphasis on them vs. the previous period.

10. Are Factors Helping in Building Better Portfolios?

We have seen enough of factors to be aware that factor is just a label for a portfolio of investable securities. In principle, the lower correlations among factors (even negative at times) vs. those among assets, seem to imply that factors offer the potential for stronger diversification. Asset classes are often positively correlated for a comparison. Nevertheless, it has been pointed out that the low correlations one observes among factors are the consequence of the fact that *replicating portfolios* contain short positions. If we allowed the same amount of short positions in portfolios built from asset classes, then the difference in the computed efficient frontier vanishes. An interesting experiment would be to compare a portfolio built with a certain number of assets to a portfolio built using, say, the first two principal components for the covariance matrix of those same assets (principal components, by construction, are uncorrelated) and allowing for the same amount of short positions in the portfolio of assets. Empirical evidence suggests that the results are very similar. Hence, some may call this supposed superiority of factors a fallacy in asset allocation.

11. Conclusion

This lesson completes the treatment of classical portfolio theory. In this lesson, we looked at three types of factors (macroeconomic, fundamental, and statistical). And, using factor models, we were able to reduce the number of parameters to estimate to perform mean-variance optimization. This holds true for any kind of optimization problem and not just Markowitz's model as long as there are parameters to estimate. However, factor models can allow us to identify a few relevant common sources of risk and can be used to create portfolios. With statistical factors, we learned to deal with factors that exist but are not directly observable. Finally, using specific indices (that to a large extent could be a result of a planned exposure to specific sources of risk/factors), we can identify portfolio managers' investment styles and assess if their styles have changed over certain periods of time. As such, factor models can become a tool of their own.

The next lesson introduces the well-known Black-Litterman model, which allows investors to introduce their feeling for markets and assets.

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